

HAMER, Norway

WMO: 013850

Lat: 60.818N

Lon: 11.069E

Elev: 222

StdP: 98.68

Time Zone: 1.00 (EUC)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.7	-16.3	-20.7	0.6	-18.4	-18.4	0.8	-16.1	7.9	0.0	6.6	0.4	1.0	60	0.589

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.8	26.5	15.9	24.2	15.6	22.1	14.8	17.7	22.8	16.9	21.4	16.0	20.2	2.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.1	11.8	19.0	15.1	11.0	18.2	14.3	10.4	17.5	50.8	23.0	48.2	21.2	45.4	20.3	22.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.9	5.8	5.0	DB	-19.8	28.8	3.1	1.7	-22.0	30.0	-23.8	31.0	-25.5	32.0	-27.8	33.2	
			WB	-20.8	19.2	3.3	1.4	-23.1	20.2	-25.1	21.0	-26.9	21.8	-29.3	22.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.3	-5.7	-4.7	-0.1	5.2	10.5	14.3	16.8	14.8	11.0	5.0	0.1	-4.0
	DBStd	8.78	5.31	5.30	4.32	3.58	4.19	3.20	3.04	2.31	2.79	3.39	4.00	5.67
	HDD10.0	2320	486	413	314	150	44	2	0	1	21	158	297	434
	HDD18.3	4783	744	646	573	394	246	128	69	111	220	413	547	692
	CDD10.0	613	0	0	0	6	59	132	211	151	51	4	0	0
	CDD18.3	34	0	0	0	0	2	7	22	2	0	0	0	0
	CDH23.3	269	0	0	0	0	24	68	170	7	0	0	0	0
	CDH26.7	41	0	0	0	0	1	7	33	0	0	0	0	0
Wind	WSAvg	2.1	1.9	1.7	2.2	2.3	2.5	2.4	2.1	2.0	2.1	2.0	2.0	1.8
Precipitation	PrecAvg	528	29	23	26	25	37	59	73	68	61	52	43	31
	PrecMax	723	71	72	70	67	76	141	166	146	117	97	77	71
	PrecMin	376	1	2	0	0	7	6	24	10	14	7	12	6
	PrecStd	92	17	14	15	17	20	36	31	39	27	28	20	17
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.9	6.3	12.0	18.6	25.9	27.7	29.4	24.3	20.7	15.7	10.6	7.6
		MCWB	3.9	3.1	6.4	10.3	14.0	15.7	17.8	16.6	13.9	12.9	8.6	5.5
	2%	DB	3.7	4.4	9.5	16.3	23.1	25.6	27.5	22.7	18.8	13.1	8.4	6.0
		MCWB	2.2	2.3	4.9	9.2	13.5	14.7	17.0	15.6	13.2	9.7	6.7	4.5
	5%	DB	2.2	3.0	7.4	14.0	20.4	22.8	25.2	21.0	17.2	11.5	7.0	4.6
		MCWB	1.0	1.4	3.5	8.1	12.6	14.2	16.4	14.8	12.8	9.2	5.7	3.0
	10%	DB	1.0	1.7	5.4	11.6	17.9	20.2	22.9	19.4	15.7	10.1	5.3	3.0
		MCWB	0.1	0.6	2.4	6.7	11.3	13.0	15.9	14.2	12.1	7.9	4.2	1.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.4	3.9	6.9	11.4	16.2	17.1	19.7	18.2	16.0	13.3	8.9	6.0
		MCDB	5.6	5.3	11.3	16.7	20.8	21.5	25.5	21.5	17.2	15.3	10.4	7.0
	2%	WB	2.7	2.7	5.2	9.8	14.8	16.1	18.2	17.3	14.7	10.9	7.1	4.7
		MCDB	3.6	4.0	8.9	14.3	20.8	22.1	24.4	20.5	17.1	12.4	8.1	5.8
	5%	WB	1.2	1.6	4.0	8.6	13.3	15.3	17.4	16.3	13.6	9.5	5.8	3.3
		MCDB	2.0	2.8	6.5	13.4	18.5	21.0	23.1	19.3	16.4	11.0	6.8	4.4
	10%	WB	0.3	0.8	2.8	7.2	12.0	14.2	16.5	15.3	12.5	8.2	4.4	1.8
		MCDB	1.0	1.8	5.1	11.4	16.7	19.3	21.5	18.0	15.1	9.7	5.2	2.8
Mean Daily Temperature Range	5% DB	MDBR	4.6	5.2	7.2	8.8	8.9	8.8	8.8	8.0	7.6	5.7	4.3	4.8
		MCDBR	5.3	6.5	9.5	12.3	12.7	11.9	11.9	10.3	9.8	7.0	6.2	6.3
		MCWBR	4.5	4.7	6.1	6.7	5.6	5.4	4.6	5.0	5.7	5.1	5.1	5.4
	5% WB	MCDBR	5.1	5.5	9.0	11.3	10.6	10.1	10.3	7.9	8.4	6.0	5.7	6.0
		MCWBR	4.6	4.2	6.0	6.3	5.1	5.1	4.6	4.5	5.3	5.0	5.0	5.4
Clear-Sky Solar Irradiance	taub	0.231	0.277	0.306	0.346	0.349	0.343	0.361	0.352	0.321	0.306	0.250	0.199	
	taud	2.178	2.281	2.304	2.342	2.438	2.483	2.445	2.462	2.536	2.492	2.263	2.064	
	Ebn at Noon	538	722	819	849	876	888	859	837	809	694	510	404	
	Edh at Noon	37	63	87	102	100	97	99	90	70	53	34	24	
All-Sky Solar Radiation	RadAvg	0.27	0.92	2.42	3.93	4.75	5.37	4.84	3.77	2.47	1.10	0.35	0.17	
	RadStd	0.05	0.18	0.24	0.46	0.48	0.46	0.63	0.34	0.30	0.21	0.09	0.01	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (55 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page